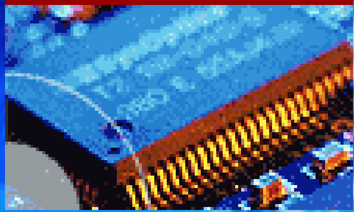
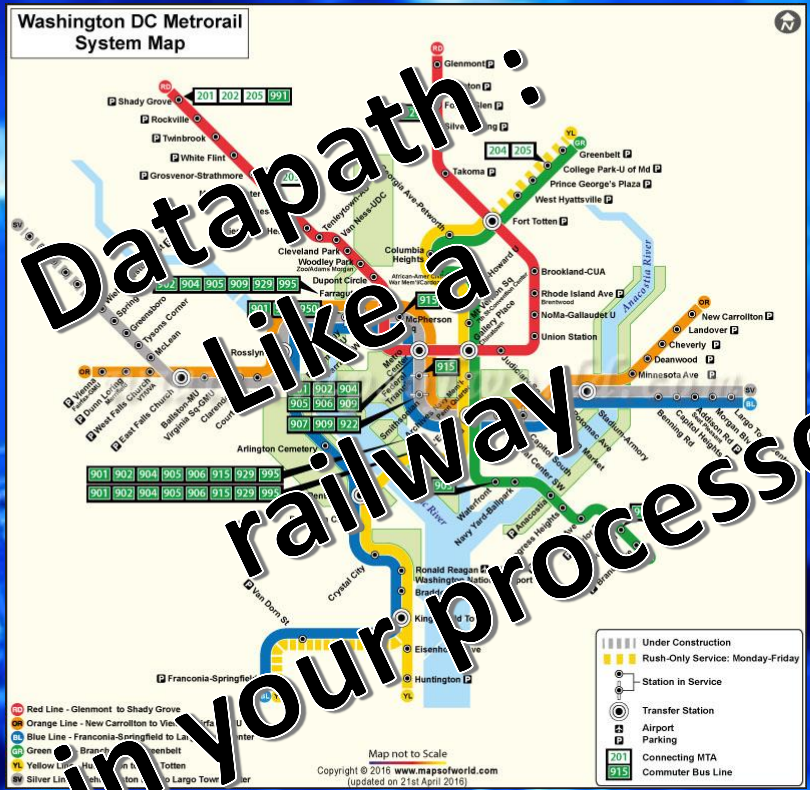
EEE3096S



Embedded Systems II



Datapath : Like a railway in your processor

CPU Architecture & Datapath

Part1/2

Embedded Systems II

L18

Dr Simon Winberg

Electrical Engineering
University of Cape Town

Outline of Lecture

- Practical Examination – What it involves
- Terminology moment: Datapath
- Computer Design Fundamentals
- Explaining the digital logic

Linked reading, for going deeper into these topics that are more introduced in these lectures:
Chapter 2 of Textbook: D. Patterson and J. Hennessy - Computer Organization and Design - ARM Edition.pdf

Terminology Moment

Temporal and Spatial Computation

Temporal Computation

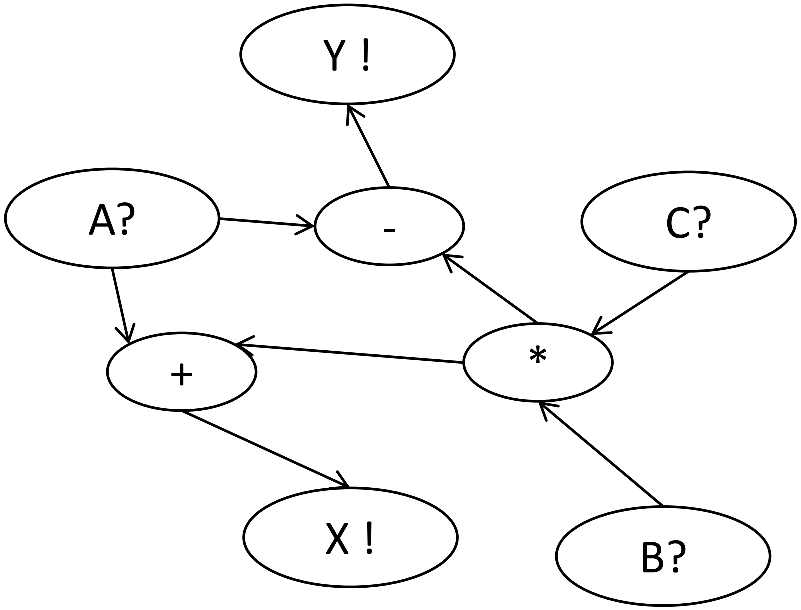
The traditional paradigm
Typical of Programmers
Things done over time steps

```
A = input("A= ? ");
B = input("B =? ");
C = input("B multiplier ?");
X = A + B * C
Y = A – B * C
```

Which do you think is easier to make sense of?

Spatial Computation

Suited to hardware
Possibly more intuitive?
Things related in a space



Can provide a clearer indication of relative dependencies.

Datapath

Datapath:
An essential concept
For processor (and
Computer) design

These slides relate to:
Course Notes Section 2.5 Computer Architecture

Datapath: Basic Concept … metaphorically

Can thing of a datapath as something like a railway (or underground) transport system).

This is where one has schedules, ways to move people (i.e. ‘data’), ways that people are admitted to the network, board the ‘path’, time it takes to get some one place to the other (transport happening at similar speed).

Datapath

- Definition: data path or **datapath**
  - This is the set of functional units that carry out data processing operations for a computer system.
  - The datapaths, together with a control unit and ALU, makes up the CPU of a computer
  - Larger datapath (or composite datapaths) can be created by joining more than one together using (e.g.) multiplexers
- Reconfigurable datapaths *:
  - These are datapaths that can be re-purposed at run-time using a programmable fabric e.g. may allow for more efficient processing and substantial power savings for particular types of application

* Source: https://en.wikipedia.org/wiki/Datapath

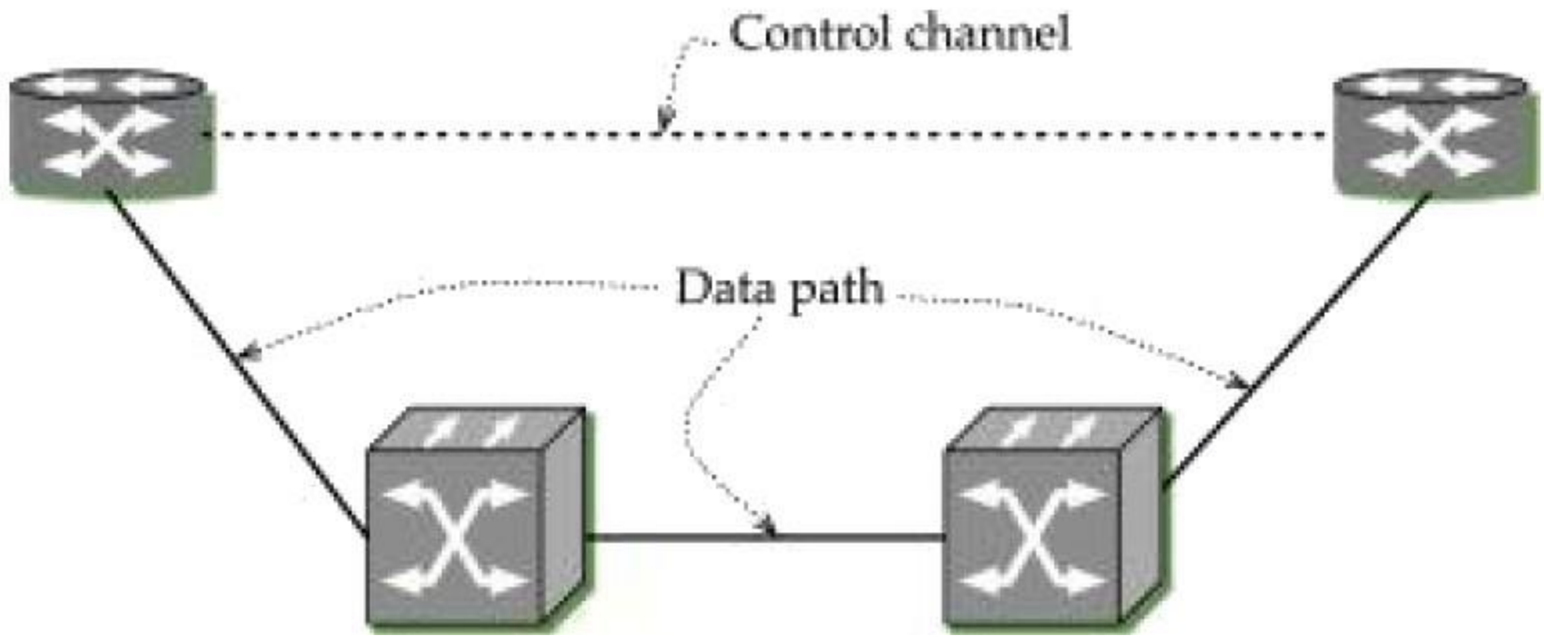
Computer Design Fundamentals

Datapath & Control

- A processor design can be considered as comprising
  - *Datapath:* moving data/signals around; and
  - *Control:* making decisions (e.g. whether or not to do an operation) and doing operations (e.g. adding two registers)

See: Course Notes Section 2.5 Computer Architecture

Datapath & Control : Basis of a processor



The control channel may be separate to the datapath, for example the signals to request to send data may be distant from the channel that transfers the data.

Defining datapath

- Definition: data path or datapath =

  The set of functional units that carry out data processing operations for a computer system
- The datapaths, together with a control unit and ALU, makes up the CPU of a computer
- Larger datapath (or composite datapaths) can be created by joining multiple datapaths together using (e.g.) multiplexers*

* This approach of composite datapaths is the commonly used practice for modular integration that is used for FPGA-based or VLSI design, e.g. for System on Chip (SoC).

Ref source / further reading: https://en.wikipedia.org/wiki/Datapath

Reconfigurable datapaths

supplementary

- Datapaths are not always static
- Reconfigurable datapaths =

  Datapaths that can be re-purposed at run-time using a programmable fabric e.g. may allow for more efficient processing and substantial power savings for particular types of application.
- Reconfigurable datapaths are nowadays generally found only for FPGA-based designs. There are some examples of processor chips that had or almost had these datapaths…
  - The VECTIS Dataflow Processing Units (DPUs) allowed for adjustable datapaths in parts of its CPU, cache and memory integration. (But these are not longer made).
  - The Power PC (use in the Sony PlayStation 3) allowed configuration of memory access – but this was not exactly a reconfigurable datapath, more option of static paths that could be selected.

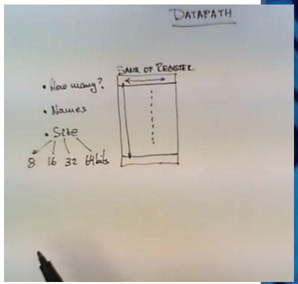
Approach to Computer Design

- The specification of a computer is provided by defining its appearance to the programmer at its lowest level, its Instruction Set Architecture (ISA) level
- From the ISA the computer architecture is developed…
- Computer architecture development essentially involves deciding its datapath and control.
- Is also an effective approach for designing a processor/CPU or designing a special-purpose application accelerator or co-processor

Note, and remember slide on spatial computation, that you are not limited to using instructions as the means to design a computer; but it is the regular practice used for computer design. Also, the instructions do not necessarily have to run in sequence they could be parallel or collaborative/interdependent.

But what exactly is the Datapath?

- Datapaths
  - Most generally this refers to the registers, processing units, and interconnections (busses) that are used to process and transfer data in a computer system
- Datapath comprises
  - A set of registers (that store data)
  - Microoperations to perform operations on data stored in the registers
  - Control interfaces (for sequencing and arbitrating operations)



Recommended video: "How a datapath works inside a computer system". Available at: https://youtu.be/ibYYqvp9FmU

Let’s design a Computer Processor!

(we will design just a simple processor of course, which builds on your understanding of digital logic done previously; but I’ll also provide some reminders and explanation of the parts used)

A simple 4-register RISC Harvard CPU

- Reminder:
  - von Neumann architecture: uses the same memory for both instructions and data
  - Harvard: has separate memory for instructions and for data
- Harvard is often preferred for embedded systems as it can be a more robust and secure having instructions and data separate (but changing the program instruction memory while running may be blocked)

We're going to use a Harvard design for our processor

Instruction Set Architecture (ISA) Planning

- We’re going to do a 4-Register RISC processor design (code named 4RR *pronounced foorrrrr*)
- ISA for our 4RR
  - Basic set of operations (e.g. ADD, AND, JMP, MOV, NOT, OR, SUM)
  - Syntax: <OP> R_{dest}, R_{s1}, R_{s2}
  - E.g.: ADD r1, R2, R3 // R1 = R2 + R3

<OP> above is the name for an ‘Op Code’ or ‘Operation Code’ e.g. 0010_2 -> ADD

Don’t worry too much about the assembly, yet. We will get to that later. We cover just some basics of instructions and assembly first, and only later in term 2 do we get to specifics of actual ARM instructions and using ARM assembly.

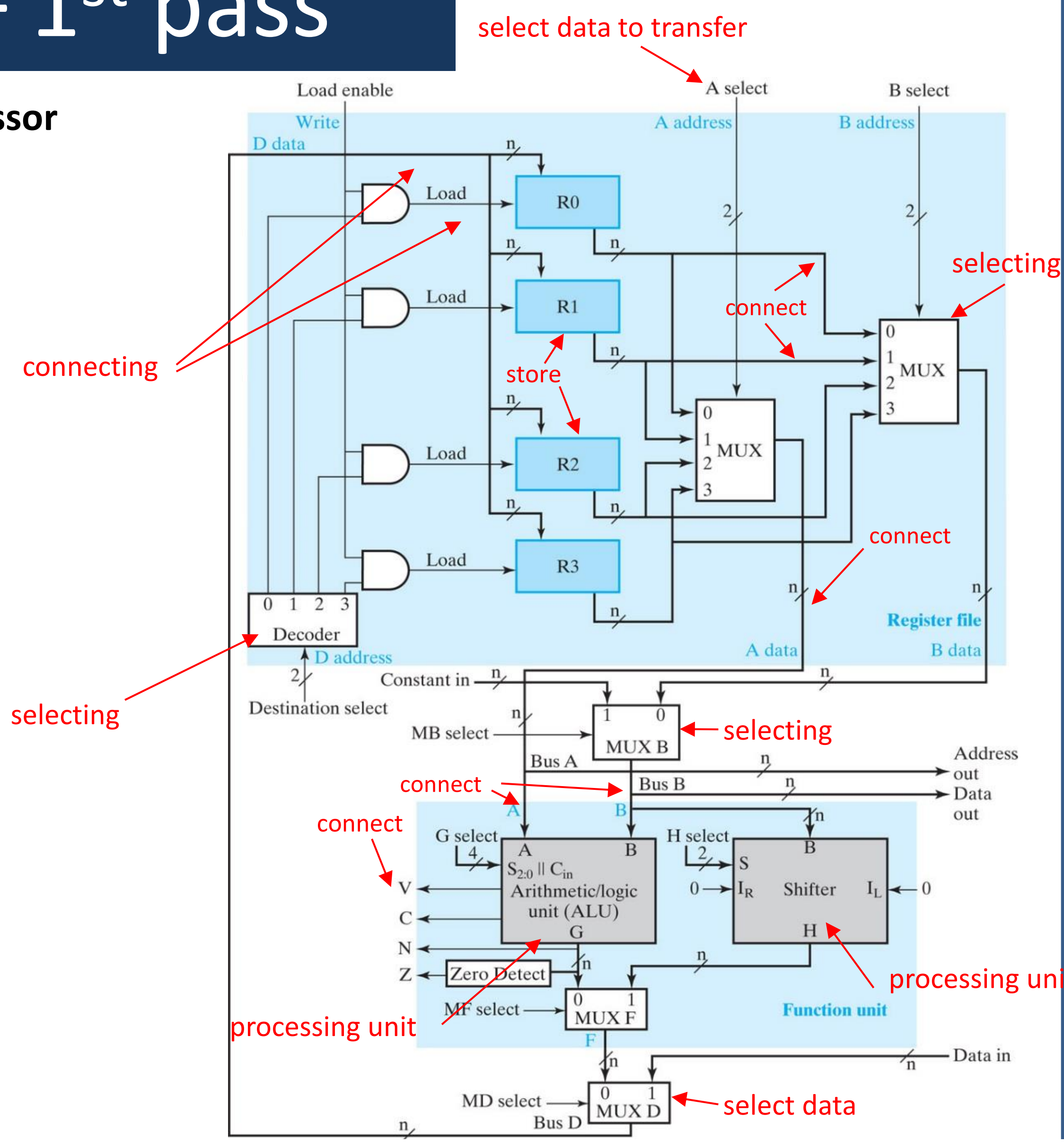
Datapath – 1st pass

The 4-Register RISC “4RR” Processor

Datapath design is largely about the circuitry to <u>store</u> data and to <u>select</u> and <u>connect</u> data to be transferred between <u>processing</u> units in a processor

A briefly run through the parts…

Onwards to….



NB: will recap the digital logic designs of the parts next

Example of a Datapath

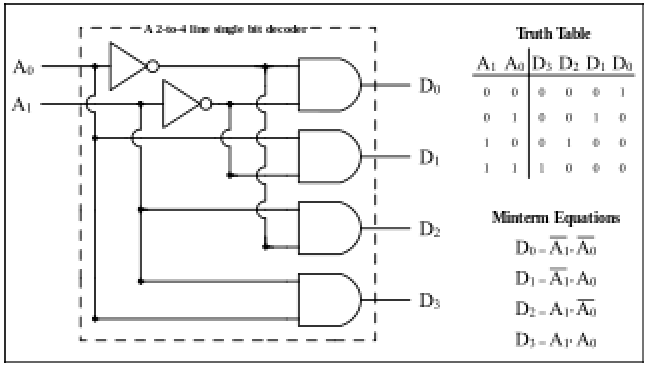
Based on CH8 Fig. 8-1 of *Logic and Computer Design Fundamentals*, Fifth Edition, by Mano, Kime & Martin. Pearson. 2016.

Brief Refresher …

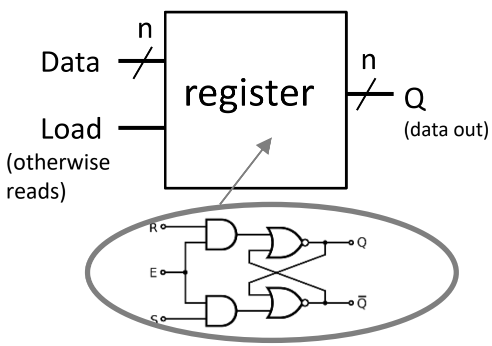
- Main digital components used:
  - Decoder
  - Register
  - Multiplexer
- We will quickly go through each of these with examples connected to the 4RR processor

Brief refresher – processor pieces

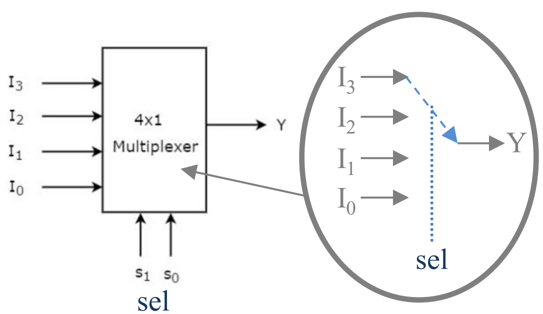
Ahhhh!



The Decoder
'Fans out bits'
i.e. convers m-bit input
to activating one of 2^m output
signals. This is used for decoding an
instruction, to chip select one of
multiple operations



The Register
Stores values. Basically a D-type flip flop that
can be enabled(load)/disabled(read)

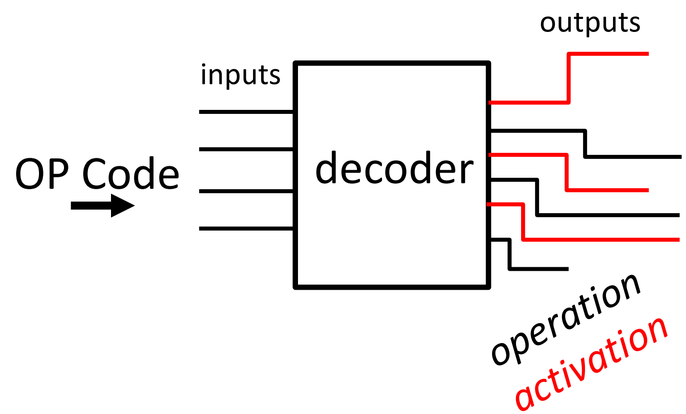


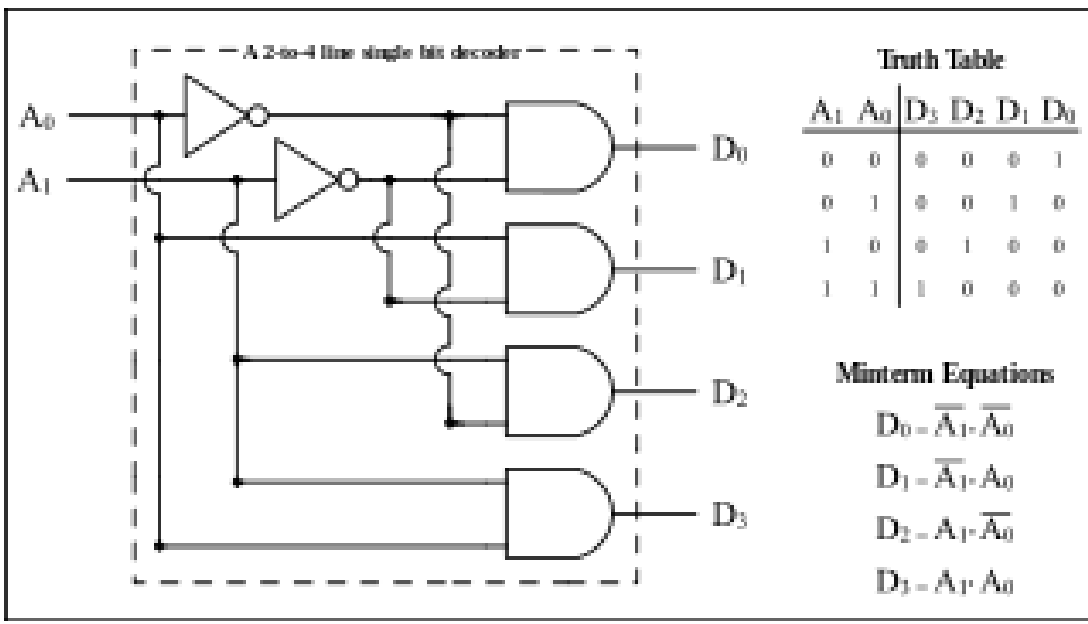
The Multiplexer
m-input sel line to select 1 of 2^m input
lines (I) to connect through to the output
line (Y)

Each explained next …

Working of The Decoder

The decoder digital logic circuit basically concerts its binary bus input to an activation of output lines. It doesn't do any operation besides setting high or low output lines for a given input. The output is undefined while the input is changing. There may be a clock line (or synchronization of output line changes) for which outputs change at a clock edge.



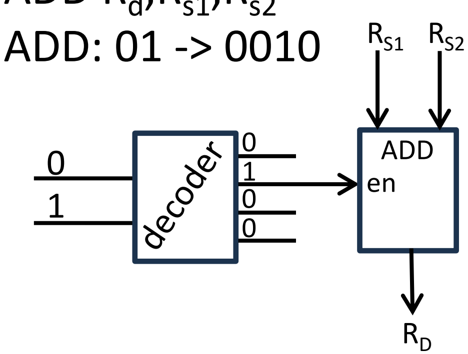


A standard decoder that simply sets output bit A_0A_1 high and all other bits low.

e.g.

ADD R_d,R_{s1},R_{s2}

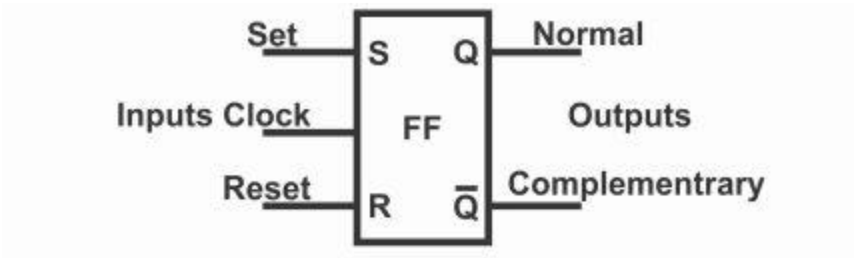
ADD: 01 -> 0010



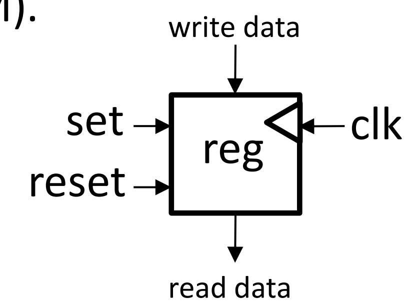
Working of The Register

The register is sent data (Data lines). A positive edge on the Load line transfers the bits on Data to the register's memory. A flip-flop provides the memory, the power has to remain on to maintain the memory (i.e., it is SRAM).

RS Flip-flop = Reset/Set flip-flop

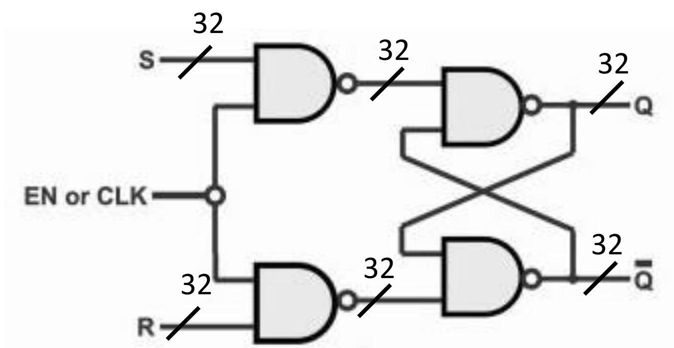


Logic Symbol for RS flip-flop



Register in ISA

Each line can be multiple bits…



RS Flip-flop digital gates circuit

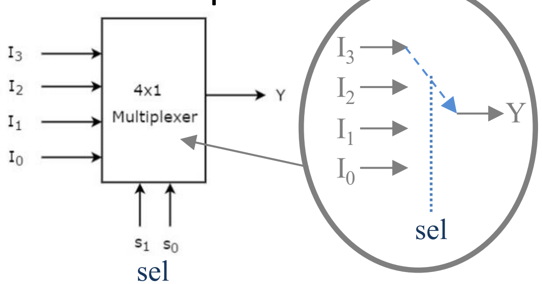
Mode of operation	INPUTS			INPUTS		
	CLK	S	R	Q	$\overline{Q}$	Effect on output Q
Hold		0	0			No change
Reset		0	1	0	1	Reset or cleared to 0
Set		1	0	1	0	Set to 1
Prohibited		1	1	1	1	Prohibited- do not use

RS Flip-flop truth table

Working of The Multiplexer

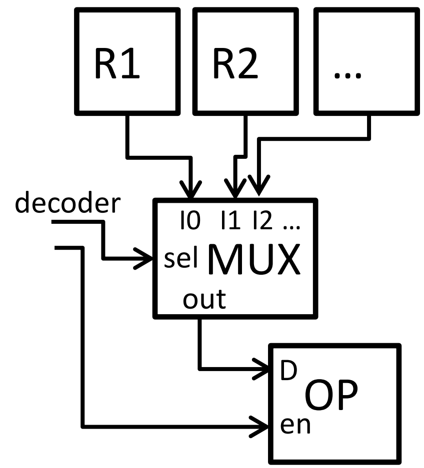
The multiplexer can be consider something of a slide-switch, where one of multiple input lines can be switched over to one output line. The input lines can be buses as apposed to just single wires. The select line ('sel' line in diagram) selects which of the inputs are forwarded to the output.

Slide-switch equivalent



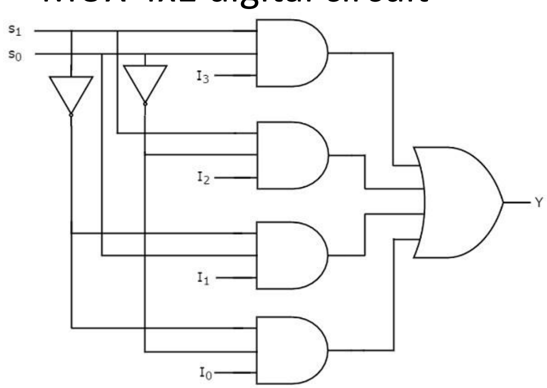
MUX4x1 Truth Table

S1	S2	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

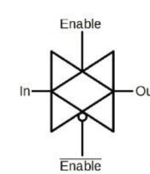


E.g.: decoder sets MUX to select register to feed into operation OP.

MUX 4x1 digital circuit



A digital circuit like that on the left would work… it is inefficient in terms of footprint, resources and power use. Instead 'transmission gates' (like transistors or NMOS or PMOS FETs) are used and activated in parallel from an enable.



transmission gate circuit symbol

The Next Episode...

Lecture L19

- Deeping into processor design & control paths, taking this lecture a bit further.